# MSG No. 133.462 $P4'_2/nb'c$ [ Type III, tetragonal ]

Table 1: Wyckoff site: `4a`, site symmetry: `22'2'.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[1,2,11,12]` |
| 2 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[3,4,9,10]` |
| 3 | $\left[\frac{3}{4}, \frac{3}{4}, 0\right]$ | `[5,6,15,16]` |
| 4 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[7,8,13,14]` |

Table 2: Wyckoff site: `4b`, site symmetry: `22'2'.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[1,2,11,12]` |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[3,4,9,10]` |
| 3 | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | `[5,6,15,16]` |
| 4 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[7,8,13,14]` |

Table 3: Wyckoff site: `4c`, site symmetry: `2.22`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | `[1,2,3,4]` |
| 2 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | `[5,6,7,8]` |
| 3 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | `[9,10,11,12]` |
| 4 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | `[13,14,15,16]` |

Table 4: Wyckoff site: `4d`, site symmetry: `-4'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | `[1,2,13,14]` |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | `[3,4,15,16]` |
| 3 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | `[5,6,9,10]` |
| 4 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | `[7,8,11,12]` |

Table 5: Wyckoff site: `8e`, site symmetry: `-1`

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, 0]$ | `[1,5]` |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | `[2,6]` |
| 3 | $\left[0, 0, \frac{1}{2}\right]$ | `[3,7]` |
| 4 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | `[4,8]` |
| 5 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | `[9,13]` |
| 6 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | `[10,14]` |
| 7 | $\left[0, \frac{1}{2}, 0\right]$ | `[11,15]` |

| | | |
|---|---|---|
| 8 | $\left[\frac{1}{2}, 0, 0\right]$ | [12,16] |

Table 6: Wyckoff site: 8f, site symmetry: 2..

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, z\right]$ | [1,2] |
| 2 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2} - z\right]$ | [3,4] |
| 3 | $\left[\frac{3}{4}, \frac{3}{4}, -z\right]$ | [5,6] |
| 4 | $\left[\frac{3}{4}, \frac{3}{4}, z + \frac{1}{2}\right]$ | [7,8] |
| 5 | $\left[\frac{1}{4}, \frac{1}{4}, z + \frac{1}{2}\right]$ | [9,10] |
| 6 | $\left[\frac{1}{4}, \frac{1}{4}, -z\right]$ | [11,12] |
| 7 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2} - z\right]$ | [13,14] |
| 8 | $\left[\frac{3}{4}, \frac{3}{4}, z\right]$ | [15,16] |

Table 7: Wyckoff site: 8g, site symmetry: 2..

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4}, \frac{1}{4}, z\right]$ | [1,2] |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2} - z\right]$ | [3,4] |
| 3 | $\left[\frac{1}{4}, \frac{3}{4}, -z\right]$ | [5,6] |
| 4 | $\left[\frac{3}{4}, \frac{1}{4}, z + \frac{1}{2}\right]$ | [7,8] |
| 5 | $\left[\frac{1}{4}, \frac{3}{4}, z + \frac{1}{2}\right]$ | [9,10] |
| 6 | $\left[\frac{3}{4}, \frac{1}{4}, -z\right]$ | [11,12] |
| 7 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2} - z\right]$ | [13,14] |
| 8 | $\left[\frac{1}{4}, \frac{3}{4}, z\right]$ | [15,16] |

Table 8: Wyckoff site: 8h, site symmetry: .2'.

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x, \frac{1}{4}, 0\right]$ | [1,11] |
| 2 | $\left[\frac{1}{2} - x, \frac{1}{4}, 0\right]$ | [2,12] |
| 3 | $\left[\frac{1}{4}, x, \frac{1}{2}\right]$ | [3,9] |
| 4 | $\left[\frac{1}{4}, \frac{1}{2} - x, \frac{1}{2}\right]$ | [4,10] |
| 5 | $\left[-x, \frac{3}{4}, 0\right]$ | [5,15] |
| 6 | $\left[x + \frac{1}{2}, \frac{3}{4}, 0\right]$ | [6,16] |
| 7 | $\left[\frac{3}{4}, -x, \frac{1}{2}\right]$ | [7,13] |
| 8 | $\left[\frac{3}{4}, x + \frac{1}{2}, \frac{1}{2}\right]$ | [8,14] |

Table 9: Wyckoff site: 8i, site symmetry: .2'.

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x, \frac{1}{4}, \frac{1}{2}\right]$ | [1,11] |
| 2 | $\left[\frac{1}{2} - x, \frac{1}{4}, \frac{1}{2}\right]$ | [2,12] |
| 3 | $\left[\frac{1}{4}, x, 0\right]$ | [3,9] |

*continued ...*

Table 9

| No. | position | mapping |
|---|---|---|
| 4 | $\left[\frac{1}{4},\ \frac{1}{2}-x,\ 0\right]$ | [4,10] |
| 5 | $\left[-x,\ \frac{3}{4},\ \frac{1}{2}\right]$ | [5,15] |
| 6 | $\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}\right]$ | [6,16] |
| 7 | $\left[\frac{3}{4},\ -x,\ 0\right]$ | [7,13] |
| 8 | $\left[\frac{3}{4},\ x+\frac{1}{2},\ 0\right]$ | [8,14] |

Table 10: Wyckoff site: 8j, site symmetry: ..2

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ x,\ \frac{1}{4}\right]$ | [1,3] |
| 2 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$ | [2,4] |
| 3 | $\left[-x,\ -x,\ \frac{3}{4}\right]$ | [5,7] |
| 4 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$ | [6,8] |
| 5 | $\left[\frac{1}{2}-x,\ x,\ \frac{3}{4}\right]$ | [9,12] |
| 6 | $\left[x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$ | [10,11] |
| 7 | $\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$ | [13,16] |
| 8 | $\left[-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$ | [14,15] |

Table 11: Wyckoff site: 16k, site symmetry: 1

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | [1] |
| 2 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$ | [2] |
| 3 | $\left[y,\ x,\ \frac{1}{2}-z\right]$ | [3] |
| 4 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$ | [4] |
| 5 | $[-x,\ -y,\ -z]$ | [5] |
| 6 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$ | [6] |
| 7 | $\left[-y,\ -x,\ z+\frac{1}{2}\right]$ | [7] |
| 8 | $\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$ | [8] |
| 9 | $\left[\frac{1}{2}-y,\ x,\ z+\frac{1}{2}\right]$ | [9] |
| 10 | $\left[y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$ | [10] |
| 11 | $\left[x,\ \frac{1}{2}-y,\ -z\right]$ | [11] |
| 12 | $\left[\frac{1}{2}-x,\ y,\ -z\right]$ | [12] |
| 13 | $\left[y+\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$ | [13] |
| 14 | $\left[-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [14] |
| 15 | $\left[-x,\ y+\frac{1}{2},\ z\right]$ | [15] |
| 16 | $\left[x+\frac{1}{2},\ -y,\ z\right]$ | [16] |